Phase 2 Report: Data Acquisition & Setup

Group ID: F26-02 | **Track:** Track A
Repository: [https://github.com/ahmykhan/WESAD-Stress-Classification]

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Abstract

This report outlines the data acquisition, ingestion pipeline, preprocessing methodology, and initial exploratory data analysis for Phase 2. Utilizing the multimodal WESAD benchmark dataset, high-frequency continuous physiological telemetry (ECG and EDA) from 15 viable subjects was windowed and transformed into statistical feature vectors optimized for machine learning classifiers.

1 Dataset Source & Description

The project utilizes the **WESAD** (Wearable Stress and Affect Detection) dataset, a publicly available benchmark dataset designed for physiological stress and affect classification.

- **Subjects:** The dataset includes recordings from 17 subjects. Subjects S1 and S12 were discarded by the original study authors due to sensor hardware malfunctions, leaving 15 viable subjects (S2–S17).
- **Modality & Hardware:** Data was captured using chest-worn RespiBAN devices and wrist-worn Empatica E4 devices. For this phase, our pipeline strictly targets the chest-worn RespiBAN sensors, specifically the Electrocardiogram (ECG) and Electrodermal Activity (EDA) channels.
- **Sampling Rate:** Both chest modalities (ECG and EDA) were sampled at a high frequency of 700 Hz.
- **Ground Truth Labels:** The dataset provides synchronized protocol conditions sampled at 700 Hz, where Label 1 represents a Baseline/Relaxed state and Label 2 represents an acute Stress state.

2 Data Collection Method

Rather than manually downloading and aligning disparate raw files, an automated ingestion pipeline was developed in Google Colab. The raw source data was maintained in Google Drive as synchronized `.pkl` (Python Pickle) binary dictionary structures.

The automated ingestion pipeline performs the following sequence:

1. Programmatically iterates through all 15 valid subject archives (`S2.pkl` through `S17.pkl`).
2. Unpacks the hierarchical dictionary structure to extract `data['signal']['chest']['ECG']` and `data['signal']['chest']['EDA']`.
3. Extracts the synchronized `data['label']` ground-truth vector to ensure time-series telemetry accurately reflects experimental protocol states.

3 Preprocessing & Feature Engineering

The raw 700 Hz telemetry yielded approximately 3,000,000 observations per subject, creating computational bottlenecks for baseline model training. The following preprocessing steps were applied:

- **Label Filtering:** Transient setup periods (Label 0), amusement intervals (Label 3), and meditation intervals (Label 4) were removed. The dataset was isolated strictly to binary target states: Baseline/Relaxed (Label 1) and Stressed (Label 2).
- **Temporal Windowing:** Continuous 700 Hz streams were partitioned into non-overlapping 5-second windows. At 700 Hz, each window spans exactly $N = 3{,}500$ discrete readings.
- **Statistical Feature Extraction:** For each window segment, raw biological readings were reduced into four summary statistics for both ECG and EDA:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i, \quad \sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu)^2}, \quad x_{\max} = \max(x), \quad x_{\min} = \min(x) \tag{1}$$

- **Target Mapping:** A singular target label was assigned to each window by calculating the statistical mode of the corresponding ground-truth time series slice.

4 Brief EDA (Key Insights)

- **Dimensionality Reduction:** The raw dataset contained over 40 million rows across 15 subjects. Applying 5-second windowing compressed the dataset to 5,504 feature vectors without discarding physiological variance.
- **Physiological Biomarkers:** Exploratory statistical checks reveal pronounced differences in `EDA_Mean` and `EDA_Std` between baseline and acute stress conditions, confirming sympathetic nervous system arousal during cognitive stress induction.
- **Data Integrity:** The final feature set contains zero null/missing values and maintains clean binary stratification across subject cohorts.

5 Final Dataset Summary

The ingestion and feature extraction pipeline outputs a master dataset named `WESAD_ML_Ready_Features.csv`, structured for direct use with Scikit-Learn, PyTorch, and TensorFlow.

- **Total Observations:** 5,504
- **Total Features:** 10

Column Name	Data Type	Description
Subject	Categorical	Subject identifier (S2–S17) for cross-validation splitting
State	Target Label	Ground-truth physiological condition (Relaxed vs. Stressed)
ECG_Mean	Float	Mean amplitude of ECG signal within the 5-second window
ECG_Std	Float	Standard deviation of ECG signal within the window
ECG_Max	Float	Maximum recorded ECG amplitude within the window
ECG_Min	Float	Minimum recorded ECG amplitude within the window
EDA_Mean	Float	Mean Electrodermal Activity within the window
EDA_Std	Float	Standard deviation of EDA within the window
EDA_Max	Float	Maximum Electrodermal Activity within the window
EDA_Min	Float	Minimum Electrodermal Activity within the window

Table 1: Feature schema of the processed dataset (WESAD_ML_Ready_Features.csv).